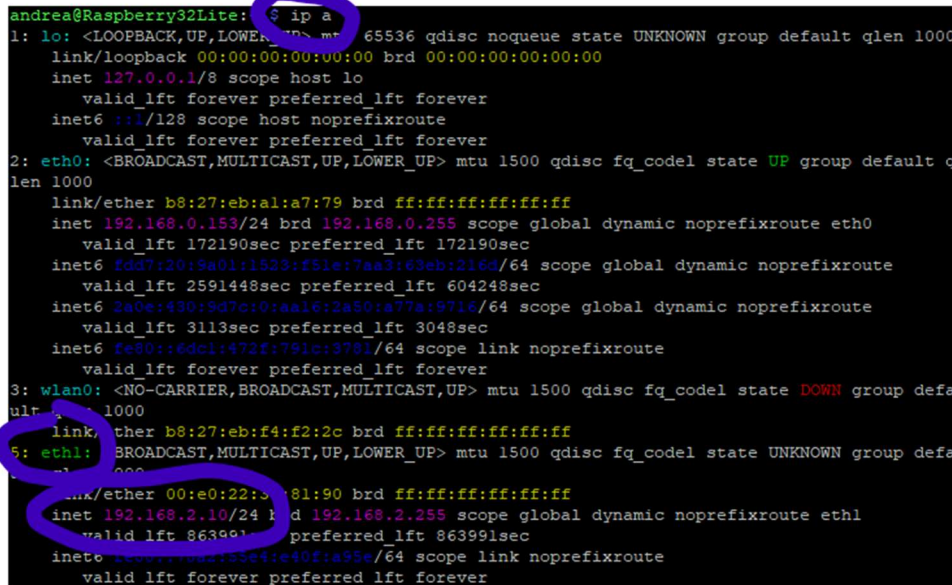


# PLUTO VIA RASPBERRY

- 1) Prepare an SD card for Raspberry Pi with OS 32bits Light using the Raspberry Imager or other system and connect the Pi to the PC via USB
- 2) Connect to the Pi via SSH (for example using Putty COMxx on port 22)
- 3) Login with username and password you defined in the Raspberry Imager
- 4) Install iptables

```
sudo apt install iptables
sudo apt install iptables-persistent -y
sudo apt install tcpdump -y      (only for tests if something goes wrong)
```

- 5) Verify the connection of the Pluto using the command "ip a"



```
andrea@Raspberry32Lite:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether b8:27:eb:a1:a7:79 brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.153/24 brd 192.168.0.255 scope global dynamic noprefixroute eth0
        valid_lft 172190sec preferred_lft 172190sec
    inet6 fd37:120:9a01:1523:f51e:7aa3:c3eb:216a/64 scope global dynamic noprefixroute
        valid_lft 2591448sec preferred_lft 604248sec
    inet6 2a0e:430:9d7c:0:aal6:2a50:a77a:9714/64 scope global dynamic noprefixroute
        valid_lft 3113sec preferred_lft 3048sec
    inet6 fe80::e6d1:472f:791a:3761/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: wlan0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default qlen 1000
    link/ether b8:27:eb:f4:f2:2c brd ff:ff:ff:ff:ff:ff
5: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    link/ether 00:e0:22:3c:81:90 brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.10/24 brd 192.168.2.255 scope global dynamic noprefixroute eth1
        valid_lft 863991sec preferred_lft 863991sec
    inet6 fe80::40d1:4354:4e40:f1a95e/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

- 6) To connect to the Pluto from the PC we have to forward what arriving on the ethernet port (for me 192.168.0.151) of the Pi to the USB port (eth1). To avoid connection conflicts, the port to connect to Pluto cannot be 22 used by the Pi but a different number (for me 23). WEW need to re-route cdata from eth0 (lan) to eth1 (USB. Unfortunately, eth0 and eth1 can swap depending on the start sequence so, before create the port forwarding, it is better to name the device Pluto with a specific name through an UDEV rule (in my case I called it pluto-usb). The rule should work with any Adalm Pluto or Plus Plus because it is recognized by vendor and product ID.

## For Raspberry with OS 32 bits

Note: nano editor can be closed with ctrl+X and the file saved following the instructions on screen.

Create a configuration file for UDEV

```
sudo nano /etc/udev/rules.d/99-pluto-net.rules
```

insert the following line

```
SUBSYSTEM=="net", ACTION=="add", ATTRS{idVendor}=="0456", ATTRS{idProduct}=="b673",
NAME="pluto-usb")
```

As alternate write in the specific mac address of your Pluto (you can read it with the command "ip link show dev eth0 or eth1). This solution is much stronger but you cannot change pluto without changing the setup.

```
SUBSYSTEM=="net", ACTION=="add", ATTR{address}=="00:e0:22:3c:81:90", NAME="pluto-usb"
```

## Per Raspberry OS 64 bits (untested!!!)

Create a configuration file

```
sudo nano /etc/NetworkManager/conf.d/10-pluto-usb.conf
```

insert the following lines

```
[device-pluto-usb]
```

```
match-device=match-property:ID_VENDOR_ID==0456,match-property:ID_MODEL_ID==b673
```

```
set-name=pluto-usb
```

Disconnect and reconnect UDEV

```
sudo udevadm control --reload-rules sudo udevadm trigger --attr-match=subsystem=net
```

Restart the NetworkManager

```
sudo systemctl restart NetworkManager
```

check that it works

```
ip a
```

if not restart the Pi.

```
andrea@Raspberry32:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth-pluto: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UNKNOWN group default qlen 1000
    link/ether 00:e0:22:3c:81:90 brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.10/24 brd 192.168.2.255 scope global dynamic noprefixroute eth-pluto
        valid_lft 863908sec preferred_lft 863908sec
    inet6 fe80::4c68:3695:5750:a1fa/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether b8:27:eb:a1:a7:79 brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.153/24 brd 192.168.0.255 scope global dynamic noprefixroute eth1
        valid_lft 172713sec preferred_lft 172713sec
    inet6 fdd7:20:9a01:1523:7fe6:d49d:445d:c1dd/64 scope global dynamic noprefixroute
        valid_lft 2591733sec preferred_lft 604533sec
    inet6 2a0e:430:9d7c:0:b032:72ef:30f4:7b14/64 scope global dynamic noprefixroute
        valid_lft 3333sec preferred_lft 3333sec
    inet6 fe80::fadf:c77e:f55d:f803/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
4: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether b8:27:eb:f4:f2:2c brd ff:ff:ff:ff:ff:ff
    inet 192.168.0.16/24 brd 192.168.0.255 scope global dynamic noprefixroute wlan0
        valid_lft 172711sec preferred_lft 172711sec
    inet6 2a0e:430:9d7c:0:ba27:ebff:fef4:f22c/64 scope global dynamic mngtmpaddr proto kernel_r
        valid_lft 3333sec preferred_lft 3333sec
    inet6 fdd7:20:9a01:1523:ba27:ebff:fef4:f22c/64 scope global dynamic mngtmpaddr proto kernel_r
        valid_lft 2591733sec preferred_lft 604533sec
    inet6 fe80::ba27:ebff:fef4:f22c/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
andrea@Raspberry32:~$
```

(eth-pluto now pluto-usb to avoid naming convention problems)

## Setup the Port forwarding

Create a configuration file read at the startup

```
sudo nano /etc/sysctl.d/99-ip-forward.conf
```

with this line

```
net.ipv4.ip_forward=1
```

Clean the NAT table

```
sudo iptables -t nat -F
```

# Port Forwarding (Porta 23 SSH, Porta 80 Web, Porta 30431 SDR II0)

```
sudo iptables -t nat -A PREROUTING -p tcp --dport 23 -j DNAT --to-destination  
192.168.2.1:22
```

```
sudo iptables -t nat -A PREROUTING -p tcp --dport 80 -j DNAT --to-destination  
192.168.2.1:80
```

```
sudo iptables -t nat -A PREROUTING -p tcp --dport 30431 -j DNAT --to-destination  
192.168.2.1:30431
```

# Masquerade verso l'interfaccia fissa del Pluto

```
sudo iptables -t nat -A POSTROUTING -o pluto-usb -j MASQUERADE
```

Rewrite the routing table

```
sudo iptables-save | sudo tee /etc/iptables/rules.v4
```

That's it, at restart you can test the connection with the Pluto

1. via SSH with the same IP address of the Pi but using port 23
2. via a web browser using the same ip address on the standard port 80
3. with SDR Console using the standard port 30431
4. any other system...

Other ports to other devices can be forwarded using the same system.

GL and 73 by I2TPY